

Quantum Machine Learning

Lab 1: Quantum States & Measurement using Qiskit

Hilbert Space, Tensor Products, Measurement, and Entanglement

Objective

In this lab, you will use Qiskit to represent quantum states, compute measurement probabilities, simulate repeated measurements, build tensor-product states, and observe the difference between separable and entangled two-qubit states.

Software Requirement

Use a Python environment with Jupyter Notebook or JupyterLab. The notebook for this lab includes an installation cell. If packages are missing, run that cell first.

```
pip install "qiskit[visualization]" ipywidgets matplotlib numpy pylatexenc
```

Concepts Used in This Lab

Concept	Meaning in this lab
Hilbert space	The complex vector space in which quantum states live. A one-qubit state is a two-dimensional complex vector.
Normalization	A valid state must satisfy total probability equal to 1.
Inner product	Measures overlap between two states.
Tensor product	Combines individual qubit states into a multi-qubit state.
Measurement	Converts a quantum state into a classical outcome according to squared amplitudes.
Entanglement	A multi-qubit state that cannot be written as a simple tensor product of separate qubit states.

Experiment Overview

No.	Experiment	Main takeaway
1	Basis states in Hilbert space	Qubits are vectors, not just classical bits.
2	Valid and invalid quantum states	Valid quantum states must be normalized.
3	Inner product and overlap	Inner products quantify similarity between states.
4	Tensor products	Multi-qubit systems live in larger vector spaces.
5	Measurement probabilities	Probabilities are squared magnitudes of amplitudes.
6	Sampling and shot noise	Repeated measurements reveal the underlying distribution.
7	Separable two-qubit states	Some two-qubit states are just products of one-qubit states.
8	Bell state construction	H followed by CNOT creates a maximally entangled state.
9	Entanglement correlations	Bell-state measurements show strong correlations.
10	Product state vs entangled state	Superposition and entanglement are different ideas.

Experiment 0: Environment Setup

Task

Open the supplied Jupyter notebook and run the setup/import cells.

What to check

- The imports complete without error.
- The notebook can display matplotlib plots.
- Qiskit circuit diagrams appear as either text or graphical drawings.

Takeaway

The same notebook can be run locally or in a cloud notebook environment as long as Qiskit and its visualization extras are installed.

Experiment 1: Basis States in Hilbert Space

Theory

A one-qubit computational basis has two basis states:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A general one-qubit state is

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}.$$

Procedure

1. Create the vectors $|0\rangle$ and $|1\rangle$ using NumPy.
2. Print the vectors and their labels.
3. Use Qiskit `Statevector` to display the same states.

Expected output

$|0\rangle$ has amplitude 1 for outcome 0 and 0 for outcome 1. $|1\rangle$ has amplitude 0 for outcome 0 and 1 for outcome 1.

Takeaway

A qubit is represented as a vector in a two-dimensional complex Hilbert space.

Experiment 2: Valid and Invalid Quantum States

Theory

A valid quantum state must satisfy

$$|\alpha|^2 + |\beta|^2 = 1.$$

The measurement probabilities are

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

Procedure

1. Test the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
2. Test the vector $[1, 1]^T$.
3. Normalize the invalid vector and check the probability sum again.

Expected output

The state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is valid. The vector $[1, 1]^T$ is not valid until it is normalized.

Takeaway

Quantum states are vectors with a special constraint: the sum of squared magnitudes must be 1.

Experiment 3: Inner Product, Orthogonality, and Overlap

Theory

For complex vectors, the inner product uses conjugation:

$$\langle a|b\rangle = \sum_i a_i^* b_i.$$

The value $|\langle a|b\rangle|^2$ measures how much two states overlap.

Procedure

1. Compute $\langle 0|1\rangle$.
2. Define $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
3. Compute $\langle 0|+\rangle$ and $|\langle 0|+\rangle|^2$.

Expected output

$\langle 0|1\rangle = 0$, so $|0\rangle$ and $|1\rangle$ are orthogonal. Also, $|\langle 0|+\rangle|^2 = 0.5$.

Takeaway

Inner products are the quantum analogue of similarity or overlap.

Experiment 4: Tensor Products and Two-Qubit States**Theory**

Multi-qubit states are formed using tensor products:

$$|00\rangle = |0\rangle \otimes |0\rangle, \quad |01\rangle = |0\rangle \otimes |1\rangle.$$

For two qubits, the basis is

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

$$|0\rangle = [1, 0]^T$$

$$|1\rangle = [0, 1]^T$$

$$|0\rangle \otimes |1\rangle = |01\rangle = [0, 1, 0, 0]^T$$

Two qubits require four basis states: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

Tensor product expands the state space from two basis directions to four basis directions.

Procedure

1. Use `np.kron` to compute $|0\rangle \otimes |0\rangle$, $|0\rangle \otimes |1\rangle$, $|1\rangle \otimes |0\rangle$, and $|1\rangle \otimes |1\rangle$.
2. Confirm that each state has four amplitudes.

Expected output

Each two-qubit basis state is represented by a four-dimensional vector.

Takeaway

The size of Hilbert space grows exponentially: n qubits require 2^n amplitudes.

Experiment 5: Measurement Probabilities from a Statevector**Theory**

For

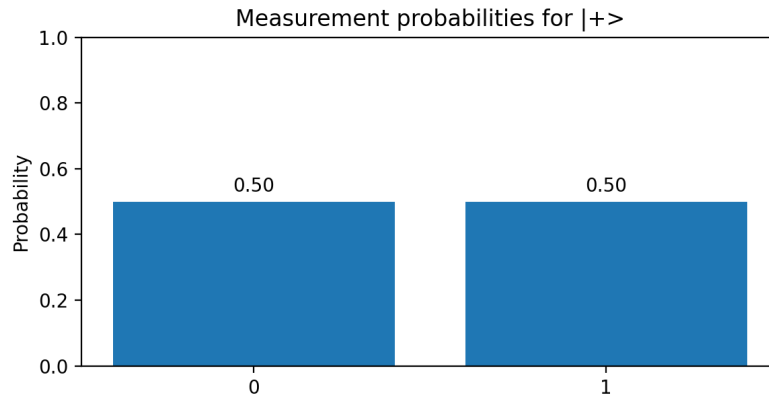
$$|\psi\rangle = \sum_i \alpha_i |i\rangle,$$

the probability of measuring outcome i is

$$P(i) = |\alpha_i|^2.$$

Procedure

1. Create $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
2. Use `Qiskit Statevector.probabilities_dict()`.
3. Plot the probabilities.



The state $|+\rangle$ gives equal probability for outcomes 0 and 1.

Expected output

$$P(0) = 0.5, \quad P(1) = 0.5.$$

Takeaway

Measurement probabilities come from the squared magnitudes of amplitudes, not directly from amplitudes themselves.

Experiment 6: Sampling and Shot Noise

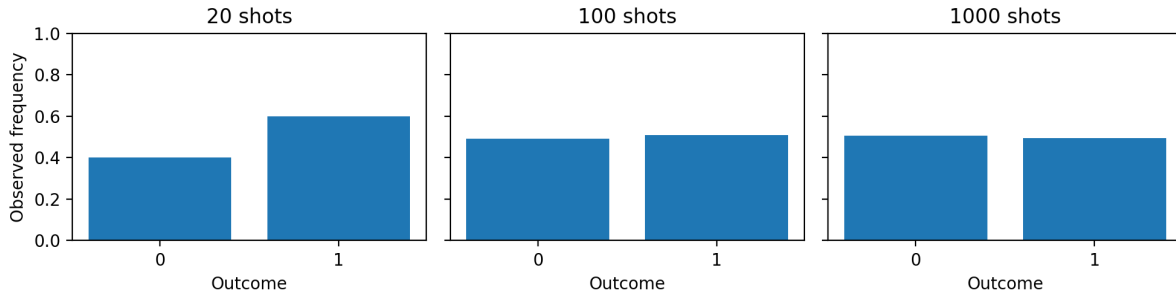
Theory

A single measurement returns one classical outcome. Repeating the same experiment many times gives a frequency distribution.

Procedure

1. Use the state $|+\rangle$.
2. Sample it with 20, 100, and 1000 shots.
3. Compare the measured frequencies with the ideal probabilities.

Shot noise: observed frequencies approach probabilities as shots increase



With more shots, observed frequencies tend to move closer to the ideal probabilities.

Expected output

The counts may not be exactly 50–50 for small shot counts. Larger shot counts should be closer to the true distribution.

Takeaway

Quantum measurement is probabilistic, and many repeated measurements are needed to estimate probabilities.

Experiment 7: Separable Two-Qubit States

Theory

A two-qubit state is separable if it can be written as a tensor product of two one-qubit states:

$$|\psi\rangle_{AB} = |a\rangle \otimes |b\rangle.$$

For example,

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Procedure

1. Prepare $|+\rangle \otimes |0\rangle$ manually using `np.kron`.
2. Prepare the same state using a Qiskit circuit.
3. Compare the nonzero amplitudes and probabilities.

Expected output

Only two of the four basis states have nonzero probabilities.

Takeaway

A separable state can contain superposition, but its qubits are still independently describable.

Experiment 8: Bell State Construction

Theory

A Bell state is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

It can be created from $|00\rangle$ using a Hadamard gate followed by a CNOT gate:

$$|00\rangle \xrightarrow{H \text{ on first qubit}} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Procedure

1. Create a two-qubit circuit.
2. Apply `h(0)`.
3. Apply `cx(0,1)`.
4. Display the circuit and the final statevector.

```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
Statevector.from_instruction(qc)
```

Expected output

The statevector has nonzero amplitudes only for $|00\rangle$ and $|11\rangle$, with equal magnitude.

Takeaway

The Hadamard gate creates superposition, and CNOT converts that superposition into entanglement.

Experiment 9: Measuring Entanglement Correlations

Procedure

1. Sample the Bell state using many shots.
2. Observe which bitstrings appear.
3. Compare the result with independent fair coins.

Expected output

Mostly 00 and 11 appear. Outcomes 01 and 10 should be absent in the ideal simulation.

Takeaway

Entanglement produces strong correlations. The two measurements are individually uncertain, but their outcomes are linked.

Experiment 10: Product State vs Entangled State

Theory

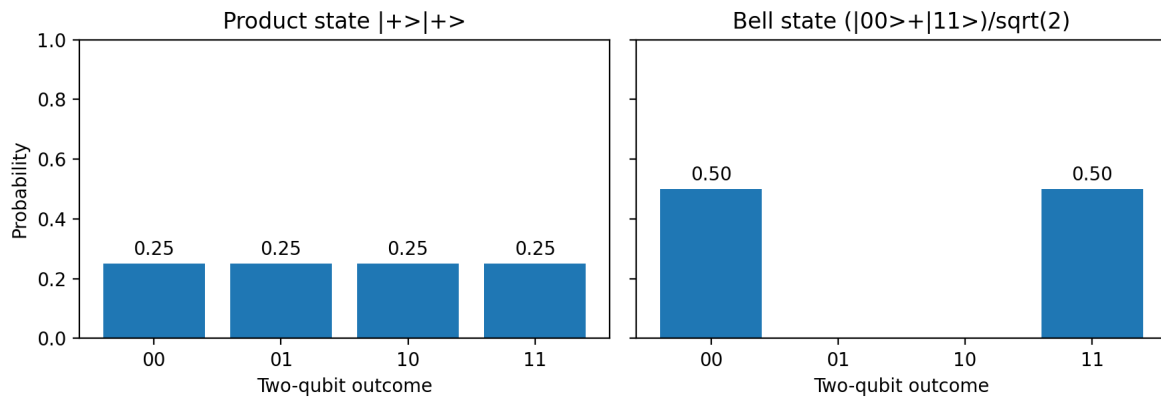
The product state

$$|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

and the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

both use superposition, but only the Bell state is entangled.



A product superposition and a Bell state have different probability patterns and different correlation structure.

Procedure

1. Prepare $|+\rangle \otimes |+\rangle$.
2. Prepare $|\Phi^+\rangle$.
3. Compare probability distributions and reduced density matrices.

Expected output

The product state has four equally likely outcomes. The Bell state has only two equally likely outcomes, 00 and 11.

Takeaway

Superposition alone does not imply entanglement. Entanglement is about non-separable correlation between subsystems.

Student Exercises

Complete these in the notebook.

1. Create the state $\frac{1+i}{2} |0\rangle + \frac{1-i}{2} |1\rangle$. Verify whether it is normalized.
2. Compute $\langle + | - \rangle$, where $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.
3. Construct $|1\rangle \otimes |+\rangle$ manually using `np.kron`. List the nonzero amplitudes.
4. Build a circuit that creates $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Verify it using the statevector.
5. Compare sampling results for 50, 500, and 5000 shots for the Bell state.
6. Explain why observing only 00 and 11 in the Bell state does not mean the system was classically predetermined.

Lab Record

For each experiment, record:

- the circuit or state prepared,
- the statevector or probability distribution,
- the sampled measurement counts where applicable,

- one sentence explaining the main observation.

Final Takeaways

- Quantum states are complex vectors in Hilbert space.
- Valid states are normalized so that total measurement probability is 1.
- Tensor products combine qubits and enlarge the state space.
- Measurement probabilities come from squared amplitudes.
- Sampling produces classical outcomes and reveals probabilities only statistically.
- Entangled states cannot be described as independent one-qubit states.
- The Bell state is the simplest practical example of entanglement in a two-qubit circuit.